University of Central Punjab

Faculty of Information Technology

Introduction To Data Science (CSDB4313) - BSCS - [redacted] 2025 - Midterm Exam

Name: _____ **Reg. No.:** _____

Objective

Time: 20 mins **Marks : 10**

Question 1: Choose the most appropriate answer and write the correct option identifier in the table given below with in the respective cell. Unreadable/overwritten answers will not be marked.

1. d	2. c	3. d	4. b	5. d
6. b	7. b	8. a	9. b	10. d

1. Which step in a data pipeline is responsible for converting raw data into a structured, usable format?
   (a) Deployment
   (b) Data Wrangling
   (c) Feature Extraction
   (d) Model Evaluation

09

2. Which Pandas function displays the first few rows of a dataset?
   (a) df.tail( )
   (b) df.describe( )
   (c) df.head( )
   (d) df.info( )

One-hot encoding is used for which type of data?
(a) Continuous Features
(b) Boolean Feature
(c) Textual Paragraphs
(d) Categorical Features

e NaN (Not a Number) values in a dataset are handled during:

) Data Visualization
Data Cleaning
Data Encoding
Feature Extraction

5. In a Confusion Matrix, "False Positive" means?

(a) Model predicted correctly
(b) Model made no prediction
(c) Model predicted negative, and it was negative
(d) Model predicted positive, but it was actually negative

6. Which function is used to create a NumPy array from a Python list?

(a) np.make()

(b) np.array()

(c) np.arrange()

(d) np.create()

7. What will be the output of the following code?

```
import seaborn as sns
import pandas  as pd
df = pd.DataFrame({'x':[1,2,3],'y':[1,2,3]})
print(df.corr())
```

A. 0 B. 1 C. -1 D. 0.5

8. What will be the output of following code?

```
import numpy as np
data = np.array([1, 2, 3, 4, 5])
print(np.mean(data))
```

A. 3 B. 5 C. 2 D. 4

9. Why is recall a critical metric in disease detection models?
(a) It ensures that no negative cases are misclassified as positive
(b) It measures the model's ability to identify all positive cases
(c) It evaluates the speed of prediction
(d) It penalizes the model for overfitting

10. Feature engineering involves:
a) Creating features for better model performance
b) modifying or transforming features for better model performance
c) Visualizing the dataset only
d) Both a & b.

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Name: M. Sufyan Abubakar Reg. No.: 0307

Time: 1 hr. 10 mins Total Marks : 50

Note: Understanding the question is part of the exam. Attempt all questions on answer sheet only.

Subjective

Note: Attempt all Questions

Question 2: **(3 x 4 = 12)**

The following data represents five years **Sales** in thousands of dollars (X) and the corresponding **Profit Earned** (Y).

Sales (X)	Profit Earned (Y)
3	4.3
5	6.7
7	7.5
9	10

Tasks:

a) Based on the data, determine whether a linear regression model is appropriate.
b) Use slope value 6.4 to find the linear regression equation.
c) Plot the initial line using the equation y = mx, where y = 6.4x
d) Calculate the Mean Square Error (MSE).

Question 3: Working with Pandas DataFrame and visualization **(6 + 6 = 12)**

(**Instruction**: Write python code in each part.)

You are given the following small dummy dataset collected from a local Hospital's survey:

Tasks:

PatientName	Age	City	Gender	B.P
Ahmed	15.0	Islamabad	Male	110
Khalid	37.0	Lahore	Female	90
Samia		Lahore	Female	87
Fareeda	32.0	Karachi	Female	
Mehmood	20.0		Male	103

Data Processing using Pandas

i) Use pandas to detect and handle missing values in the dataset. Fill missing 'Age' with the column mean, 'City' with the most frequent city, and 'B.P' with the column mean. 03

ii) Justify which method you prefer and why . 03

Feature Engineering

iii) Create a new column named Age_Group: 'Youth' if Age < 25 , 'Adult' if Age ≥ 25

iv) One-hot encode the 'City' and 'Gender' columns using pandas.

Question 4: Theoretical part (4 +4+ 6 = 14)

1. Differentiate one-hot encoding and label encoding. When would you use each?
2. Explain Normalization vs Standardization. Why are they important before training machine learning models?
3. What is outlier? How can you identify outliers? What are positive and negative outliers?

Question 5: Machine Learning Model Evaluation (8 + 4 = 12)

You trained a binary classification model to predict whether a patient has diabetes.

The confusion matrix is given below:

	Predicted Positive	Predicted Negative
Actual Positive	40 TP	10 TN
Actual Negative	5 FP	45 FN

Tasks:

a. Calculate Accuracy, Precision, Recall, and F1 Score.
b. What do you think could be the reason for getting high recall but low precision?